

Updates from an exciting year with the OCP Sustainability Project, DCF Sustainability Project & Roadmap

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Why: Responsibility & Impact Areas

Three stakeholder pressures define the mandate for every data center site.

Regulation

Rising disclosure and compliance obligations — CSRD, EU Energy Efficiency Directive, local water and emissions permits. Non-compliance limits our license to build and operate.

- **Auditable energy, water & carbon reporting**
- **Permit-ready efficiency design standards**

For Local Community

Sites compete with neighbors for power, water and land. Trust determines speed of approval for the next campus.

- **Grid & water stewardship commitments**
- **Local jobs, heat reuse & transparency**

Customers

Customers carry our footprint in their Scope 3. Verified low-carbon capacity is now a buying criterion.

- **Per-workload carbon & efficiency data**
- **Credible 24/7 clean energy pathway**

Impact areas

How: Four Pillars of Work

Each pillar delivers evidence, standards and metrics that feed a single Sustainability Guide.

01

Circularity

Reuse, refurbish and recover hardware; design-out waste across the asset lifecycle.

02

Embodied Carbon

Measure and cut carbon in concrete, steel and IT hardware from design stage onward.

03

Power Metrics & Efficiency

PUE, WUE, CUE and utilization telemetry — measured consistently across the fleet.

04

Risk & Resilience

Climate, water and grid risk assessed per site with mitigation built into design.

Outcome: an all-encompassing Sustainability Guide — one source of standards, metrics and evidence fed by all four pillars.

Special Entry Shoutout

TOPIC: Designing Data Centers for My Future: A Call to Action from the Next Generation



Submitter

Abel Brink

I am 12 years old- and I represent the generation that will live with the consequences of the decisions this industry makes today. Climate change is not abstract to me- it shapes the future I will grow up in.

Public

Sustainability as the 5th Tenant of OCP

1. Openness
2. Efficiency
3. Scale
4. Impact
5. Sustainability

- Our industry is growing as fast as possible. We must find a way to grow sustainably.
- Our customers, investors, and communities all demand that we demonstrate sustainable performance.
- It is the mission of the Open Compute Project to spur the pace of innovation in the data center industry. Sustainability must be baked in to all the ideas that OCP incubates and promulgates.
- Our mission is to encourage the entire industry to develop and operationalize sustainable, scalable computing.



Operational Climate Resilience

- The mission of the Operational Resilience (OpRes) workstream is to define the **"Physics of Failure"** for data center infrastructure.
- This workstream operates as a technical bridge, translating the climate data (Sustainability Project) into operational thresholds for infrastructure/IT (DCF Project).
- Launching with participation from Google, Meta, Microsoft, Schneider Electric, and the University Space Research Association.

Carbon Disclosure: Joint workstream with iCA



Leadership

Andrea Desimone (Schneider Electric), Miranda Gardiner (iCA), Malcolm Hegeman (Google), Lalit Joshi (Microsoft), Nalintip Suwannakarn (Meta)

2026 Updates

- **Co-Lead Expansion:** Introduced Malcolm and Nalintip as new workstream co-leads; convened workshop to align methodologies and roadmap.
- **Base Specification:** Published OCP Base Spec for Embodied Carbon Disclosure (Aug 2025) to harmonize supply chain requirements and define minimum data fields.
- **Marketplace Integration:** Integrated carbon disclosure readiness features directly into the OCP Marketplace.
- **PCR Landscape Analysis:** Completed analysis with iMasons, mapping ICT/MEP guidelines against OCP taxonomy and prioritizing gap developments.

Goal: Enabling a **future state of low-carbon procurement decisions** across the supply chain

LCA PCR Development



Leadership

Meta, Microsoft, Google, AWS, Fraunhofer Institute

(Funded by Meta/MSFT/Google/AWS. Fraunhofer is the main research / implementation driver)

Goal: Building a standardized Product Category Rule (PCR) for Devices & Datacenter Equipment.

- **Workstream format**
  - Format includes closed weekly work.
  - Regular feedback points where work will be socialized with OCP community
- **2026 Updates:**
  - **Initial Draft open for Feedback –** Reach-out to Fraunhofer Institute driving the work (Eckstein, Chris chris.eckstein@izm.fraunhofer.de)
  - **Next Steps:** Public Consultaion & approvals
  - **Targetted Timeline** – Mid 2027

CIT Circularity Sub Project



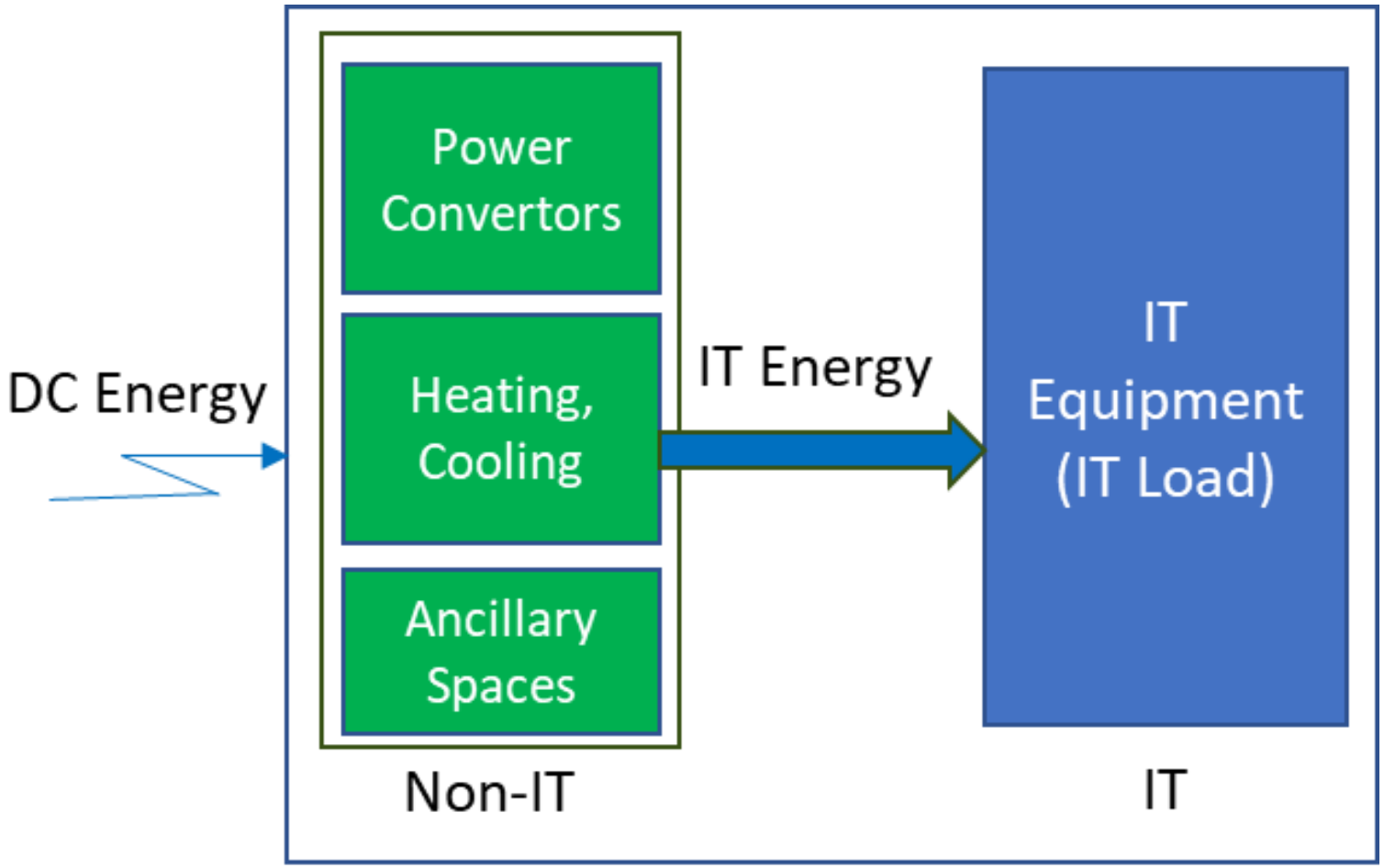
Deliverables Available:

- Circularity Guide
- Data Sanitization Guide for Circularity

Work in Progress:

- Vertical Circularity Ecosystem Whitepaper
  - Status: in progress to publish – reviews complete
- Carbon Accounting for Circularity Guide
  - In plans to Kickoff

DC Metrics / Other Power Workstreams



Improving / Establishing definitions for metrics to reflect sustainability impact

PUE
IUE
CUE

Data Center Water Stewardship



Table 1. The four linked decisions in a data center water strategy

Decision layer	Primary question	Minimum output
Regional context	Is the water source appropriate for this basin and operating profile?	Documented basin-risk and source-feasibility assessment
On-site systems	Can the facility measure, control, treat, reuse, and discharge water safely?	Water balance, metering plan, P&ID boundaries, and operating limits
Cooling selection	Which architecture best balances heat rejection, energy, water, reliability, and cost?	Climate-bin comparison and decision matrix
Accounting and disclosure	Can performance be reported consistently and assured?	Defined boundary, data-quality plan, and evidence pack

Leads:

Rishab Vardhan Harikrishnan (Oracle)

Michael Anderson (SPX)

Derek Kauneckis (Vertiv)

Megan Rehkopf (Bloom Energy)

Anastasia Behr (UL Solutions)

John Menoche (Vertiv)

Goal: Enabling a future state of low-carbon procurement decisions across the supply chain

DCF Sustainability Update

Overall Project Status

Impact on DC and Industry Sustainability with focus on Collaboration

Project Leadership Team

Priya Chhiba (Google)

Areas for Collaboration

- Net Zero Innovation Hub
- iMasons Climate Accord

Sub-Project/Workstreams

Heat / Energy / Water

Lead

Aaron Wemhoff

Carbon Intelligent LCA

Leads

Maria Abadie / Hayley Moore

Clean Backup Power

Leads

Abhik Bera / Kially Ruiz

Clean Backup Workstream

Documentation

Clean Backup White Paper
Currently being developed

Clean Backup 0.3 Spec ([Published])
V1.0 Will be published soon

Net Zero Innovation Hub Collaboration

Upcoming Panel Discussion

Testing Plan for Clean Power Options:

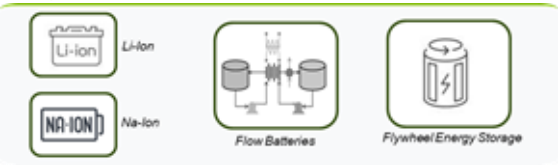
- **Internal Combustion Engine (ICE):** April 2026
- **Fuel Cells (PEMFC):** Q3-Q4 2026
- **Aluminum Air Generators (AL-Air)** - Q4, 2026/Q1, 2027

Reduce

Energy Storage: Augment existing back-up power

- Li-ion / Na-ion or alternate chemistries Batteries
- Flow Batteries
- Flywheel Energy Storage

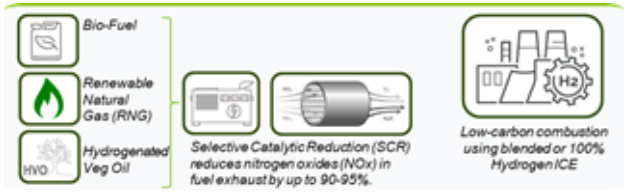
Avoid fossil combustion with onsite storage + Microgrid



Substitute

Clean Fuel and Generation: For Transition

- Bio-fuel / Renewable Natural Gas (RNG)
- Hydrogenated Veg Oil (HVO)
- Selective Catalytic Reduction (SCR)
- Low-carbon combustion (100% Hydrogen ICE)

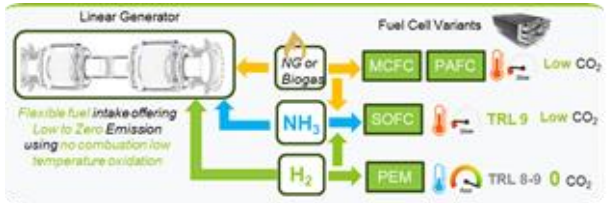


Innovate

Innovation in Zero Emission - Generation

- Linear Generators
- Fuel Cell Variants (MCFC, PAFC, SOFC, PEM)
- Al-Air Generators

Flexible fuel intake offering net zero carbon path



Intelligent Life Cycle Management (ILM)

The ILM workstream focuses on integrating **embodied carbon**, **efficiency**, and **end-of-life** considerations for holistic data center MEP equipment life cycle management.

Core Problem Statements:

Data Inconsistency

EPDs rely on subjective and variable vendor models instead of real operational data.

Lack of Comparability

PCRs are inconsistently available for complex MEP systems and equipment.

Siloed Efforts

Sustainability efforts often optimize one factor (e.g., efficiency) without evaluating trade-offs.

Cross-OCP Engagement and Collaboration

While OCP efforts establish the "what" and "how", ILM addresses the "so what"—how to actually use this data to make decisions. We will continue to actively collaborate with these workstreams to ensure cross-workstream alignment and feedback.

OCP Global Summit Presentation: Standardizing Data Center Operational Carbon Tracking: Supplier Metrics vs. Operator Needs

Low Embodied Carbon Concrete

Next Presentation: Novel Low-Carbon Materials Framework

Hyperscaler Green Concrete

Focused on industrial-scale embodied carbon reduction for data center infrastructure. (Field Pilot)



Two Learning Modules Published in the OCP Portal (Published)

Novel Materials Framework (Published)

Green Concrete Field Pilot (Published)



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Carbon Water Energy Update

Strategic Focus Areas

Balancing carbon, water, and energy in data center cooling systems.

- Incorporate life cycle impacts
- Identify critical areas for further research
- Analyze operation, geography, and waste heat recovery

Progress to Date

- **Guidelines White Paper** Published
- **Deep Dive Paper:** Examining influence of evaporative cooling on water footprints

Table 3a. Cooling Strategies: Operational Considerations

Strategy	Energy	Water (on-site, upstream, total)	Carbon (operational, inherent, total)	Notes
Evaporative cooling	D	U, D, ?	D, ?, ?	Adiabatic cooling reduces energy but adds on-site water

U: up, D: down, ?: requires further evaluation (deeper dive required)

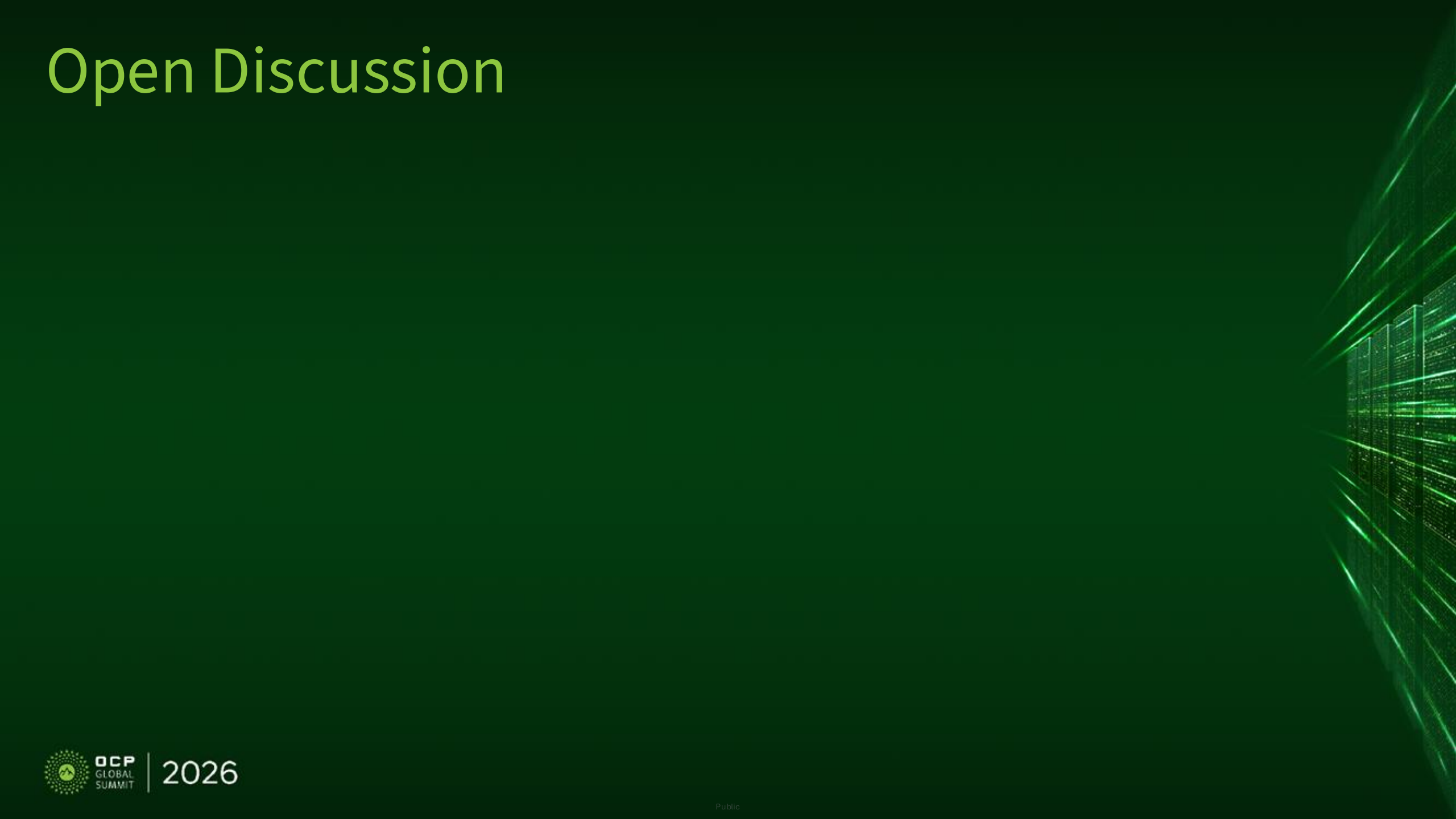
Call to Action

- Problem to solve
  - Let’s decarbonize our Data Center Infrastructure together
- How to get involved in the project/sub-project community
  - Join meetings:
    - DCF Sustainability [https://ocp-all.groups.io/g/DCF-Sustainability]
    - Sustainability Project [https://ocp-all.groups.io/g/Sustainability]
- Contributions and Future Roadmap availability
  - Clean Backup Specification (0.3) - 1.0 Targeting publication in Q4, 2026
  - Novel Materials Framework - Published
  - Green Concrete Field Pilot - Published
  - Clean Backup White Paper: 2027
  - Energy/Water/Carbon Paper: 2027





Open Discussion



Public